
Supplementary Material for “BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach”

A PROOFS AND ADDITIONAL ANALYSIS

A.1 Behavior anchor and Proposition 1

Let $b(s) = \mathbb{E}[a \mid s]$. The conditional bias-variance identity

$$\mathbb{E}[\|\mu(s) - a\|_2^2 \mid s] = \|\mu(s) - b(s)\|_2^2 + \mathbb{E}[\|a - b(s)\|_2^2 \mid s]$$

proves Eq. (7) after averaging over states and dividing by d ; its second term is independent of μ . For deterministic policies, $W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2$, so d_d^2 is a fixed rescaling of integrated squared Wasserstein distance. Dividing the normalized anchored objective $-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$ by α gives Eq. (8) with $h = \alpha/2$. More generally, $-\lambda \mathbb{E}_s \hat{Q}$ gives $h = \lambda C/2$ because $\hat{Q} = C\bar{Q}$.

A.2 Proposition 2 and critic-error transfer

By ε_k -optimality relative to the feasible comparator μ_{k-1} ,

$$\begin{aligned} & -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_k(s)) + d_d^2(\mu_k, \mu_{k-1}) \\ & \leq -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_{k-1}(s)) + \varepsilon_k. \end{aligned} \tag{1}$$

Rearranging yields Eq. (13). For $\hat{Q} = Q^\dagger + e$, substitution further gives

$$\begin{aligned} m_k^{\hat{Q}} &= \frac{2h_k}{C_k} \mathbb{E}_s [Q^\dagger(\mu_k) - Q^\dagger(\mu_{k-1})] - d_d^2(\mu_k, \mu_{k-1}) \\ & \quad + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})] \\ &= m_k^{Q^\dagger} + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})]. \end{aligned} \tag{2}$$

This proves Eq. (14). If $|e(s, a)| \leq \delta_k$ on the realized path, then

$$m_k^{Q^\dagger} \geq -\varepsilon_k - \frac{4h_k}{C_k} \delta_k. \tag{3}$$

The result concerns a trusted critic objective. A true-return statement would require additional occupancy and value-estimation assumptions.

A.3 Proof of Proposition 3

Let $F_{s'}(a) = \min_j Q_j^-(s', a)$. If every $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz, then

$$\begin{aligned} |F_{s'}(a) - F_{s'}(b)| &\leq \max_j |Q_j^-(s', a) - Q_j^-(s', b)| \\ &\leq L(s') \|a - b\|_2. \end{aligned} \tag{4}$$

Therefore,

$$|y_{\mu_1} - y_{\mu_0}| \leq \gamma L(s') \|\mu_1(s') - \mu_0(s')\|_2. \tag{5}$$

Squaring, averaging over dataset transitions, and taking the square root proves Eq. (15). Later actors do not occur because the TD target conditions only on μ_1 .

A.4 Local explicit–implicit equivalence

Let $\beta = dh/C$, $g = \nabla_a \hat{Q}(a_0)$, and suppose the action constraint is inactive. The action-metric-matched linearized update is $\Delta a_{\text{lin}} = \beta g$. A statewise exact proximal solution satisfies

$$\Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (6)$$

If the Hessian of $\hat{Q}$ is locally Lipschitz near a_0 ,

$$\nabla_a \hat{Q}(a_0 + \Delta a) = g + H\Delta a + O(\|\Delta a\|^2). \quad (7)$$

Thus

$$(I - \beta H)\Delta a_{\text{prox}} = \beta g + O(\beta \|\Delta a_{\text{prox}}\|^2). \quad (8)$$

For $\beta \|H\| < 1$ and sufficiently small β ,

$$\Delta a_{\text{prox}} = \beta g + \beta^2 Hg + O(\beta^3), \quad (9)$$

whereas $\Delta a_{\text{lin}} = \beta g$. With only C^2 regularity, the safe remainder is $o(\beta^2)$ rather than $O(\beta^3)$. Both ideal updates are first-order schemes for fixed positive C , and changing a fixed C only rescales time. A positive policy-dependent global C preserves the raw-critic ascent direction under an adaptive time parameterization in the infinitesimal limit. This derivation is not applied to the large-budget regime with active projection, ReLU-region crossings, critic drift, or inaccurate actor solves.

A.5 Proof of Proposition 4

Write $x_j = x_0 + jhf_0 + O(h^2)$ for fixed K , which follows inductively from the one-step consistency assumption. Smoothness then gives

$$f(x_j) = f_0 + jhJ_0f_0 + O(h^2). \quad (10)$$

Summing the K increments,

$$\begin{aligned} x_K - x_0 &= h \sum_{j=0}^{K-1} f(x_j) + h^2 \sum_{j=0}^{K-1} p(x_j) + O(Kh^3) \\ &= Khf_0 + h^2 \frac{K(K-1)}{2} J_0f_0 \\ &\quad + Kh^2 p_0 + O(h^3). \end{aligned} \quad (11)$$

where the final remainder uses fixed K . Substituting $h = T/K$ proves Eq. (18). The exact flow satisfies $x(T) = x_0 + Tf_0 + \frac{1}{2}T^2J_0f_0 + O(T^3)$. Thus explicit Euler has leading error $-T^2J_0f_0/(2K)$, while backward Euler has $+T^2J_0f_0/(2K)$. Both remain first-order methods.

B EXPERIMENTAL DETAILS

B.1 Algorithms and schedules

The implementation parameter is $T = \tau = \alpha/2$, and the normalized budget is split as $h = T/K$. Thus $K = 1$ recovers TD3+BC with coefficient $\alpha = 2T$. BAR-Prox uses Eq. (9) with a moving reference; BAR-Lin constructs Eq. (19) and uses MSE regression. The K actor networks are independently initialized persistent networks with separate Adam states. Each iteration first updates the online critic. On every second iteration, one optimizer step is then performed per actor in chain order on the same minibatch: actor one updates first; its target actor and the updated critic each receive a Polyak update of .005; later actors then use stop-gradient references from the updated online chain. Equations (5) and the main-text linearized scale define the actual normalization, with safeguard 10^{-6} . All methods share data, critic architecture, and 10^6 critic updates, while a K -hop run performs K actor optimizer steps and is not compute matched.

The ideal proximal map is not a certificate for this live chain. For a later persistent actor that differs from its predecessor, even $h = 0$ leaves

$$\nabla_{\theta_k} \mathbb{E}_s [\|\mu_k(s) - \text{sg } \mu_{k-1}(s)\|_2^2/d] \neq 0 \quad (12)$$

in general. One Adam step therefore need not become a near-identity map as the nominal coefficient vanishes; the ideal analysis supplies local geometry, while the sweep evaluates the implemented optimizer trajectory.

We use a 256–256 ReLU actor MLP with tanh output and twin 256–256 ReLU critic MLPs, batch size 256, Adam with learning rate 3×10^{-4} , discount .99, policy delay 2, target noise .2 and clip .5 times the action bound, and state normalization by dataset mean and standard deviation plus 10^{-3} . Every reported sweep score is the ten-episode D4RL normalized mean at the final 10^6 -update checkpoint. Archived configurations record 50,000-update intermediate evaluation for K1/P2/P3, mixed 50,000/final-only evaluation for L2, and final-only evaluation for L3. Exact historical per-run configurations for proximal seeds 2–3 were not retained; post-hoc diagnostic manifests record audit settings, not the training launch. Intermediate cadence does not enter the headline estimand. The 14-value grid spans .05 to 20. Both proximal and linearized procedures use seeds 0–3 in the headline score grids.

The nine exact datasets are the Cartesian product of domains **halfcheetah**, **hopper**, and **walker2d** with suffixes **medium-v2**, **medium-replay-v2**, and **expert-v2**. Timeout rows are omitted from transition construction; terminal flags stop bootstrapping.

B.2 Diagnostic quantities

On a fixed current-state validation batch with paired dataset actions, we compute the empirical action-averaged proxies

$$\begin{aligned}\hat{D}_{\text{critic}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_1(s) - a\|_2^2, \\ \hat{D}_{\text{final}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_K(s) - a\|_2^2.\end{aligned}$$

These quantities measure the online actor chain relative to its sample anchor. They are proxies for, rather than direct measurements of, the next-state target-policy displacement in Proposition 3: the practical TD target uses a Polyak target actor and policy-smoothing noise. By Eq. (7), the population expectation of each proxy equals squared displacement from the conditional-mean behavior map plus the same additive conditional-action variance. Population differences therefore cancel this common floor, but ratios do not: a ratio of sample-anchored proxies is not a ratio of mean-policy displacements. Finite validation batches add sampling error to both interpretations. We additionally evaluate the deterministic Polyak target actor and 64 draws of clipped target-smoothing noise on 4096 next-state transitions per checkpoint. For source index i , the audit pairs $s'_i = s_{i+1}$ with dataset action a_{i+1} ; it excludes timeout rows and excludes true terminal rows from this bootstrap-only sample. All target-exposure quantities remain squared displacement per action coordinate relative to the paired next dataset action. Thus absolute levels include conditional action variance, whereas paired differences use a common sampled anchor. For BAR-Lin, projection attenuation is the RMS norm ratio between the clipped and raw target displacements. If $\tilde{g}$ is the normalized critic action gradient and Δa is the learned actor displacement, define

$$h_{\text{eff}} := \frac{\mathbb{E} \langle \Delta a, \tilde{g} \rangle}{\mathbb{E} \|\tilde{g}\|_2^2}.$$

We report $h_{\text{eff}}/(dT/K)$ as the effective fraction of the nominal linear-target coefficient. The matched audit evaluates 512 fixed states per environment–seed run. We also record action saturation, TD-error quantiles, critic magnitude, critic loss, and gains under both the jointly trained critic and a controlled common critic: the frozen final TD3+BC $T = 1$ critic from a separate canonical reference run outside the fourteen-value sweep, matched by environment and seed. A run-total gain is the sum of mean Q changes across the K hops on this shared batch. The common critic supports cross-critic comparison but is not treated as ground truth. A run is descriptively labeled collapsed only from its external normalized return, using the threshold 20.

For the exploratory simulator-reference audit, we clone MuJoCo integration state and environment RNG from evaluation-policy trajectories, record the episode clock, and reset the TimeLimit for a continuing-task critic-calibration estimand. At each state we compare $a_1 = \mu_1(s)$ and $a_K = \mu_K(s)$ under continuations ρ_1 , μ_1 , and μ_K , where ρ_1 is the Polyak first target actor with TD3 clipped target-smoothing noise. The primary calibration target is the discounted return under this effective backup continuation. The screening covers three environments, $T \in \{4, 7, 10, 14, 20\}$, $K \in \{2, 3\}$, two seeds, and 12 states per checkpoint. It uses a 1000-step horizon, $\gamma = .99$, four common-random-number target-smoothing rollouts, and a one-return-unit ranking dead zone. It is a post-hoc screening study, not a causal routing intervention.

Table 1: Mean score over nine tasks at each total budget. Every row uses four seeds.

T	.05	.1	.2	.4	.7	1.5	2.5	4	7	10	12	14	17	20
TD3+BC	63.81	62.57	64.58	65.94	67.69	74.79	67.41	45.93	30.36	23.77	21.64	18.96	18.92	18.09
Prox-2	60.45	64.19	67.90	69.61	73.65	78.05	80.66	74.48	58.88	41.14	32.17	29.55	25.95	26.25
Prox-3	60.14	60.82	68.08	71.01	73.32	78.95	81.41	81.26	71.00	51.14	49.74	41.68	32.75	35.83
Prox-4	59.59	63.95	68.71	71.26	73.26	77.70	81.25	82.06	81.79	74.35	67.88	54.28	52.14	35.29
Lin-2	62.28	63.45	66.62	71.63	70.98	75.13	78.43	75.16	51.49	37.81	26.03	27.46	27.50	23.55
Lin-3	59.27	62.86	64.99	70.90	74.13	77.74	78.20	78.16	67.80	45.60	39.96	41.69	26.28	25.59

The route-only intervention trains two actor-detached shadow critics from identical initial parameters on the same minibatches and target-smoothing draws while the primary first-hop critic continues to drive the shared actor trajectory. The four environment–budget cells—Hopper-medium at $T \in \{10, 20\}$ and Walker2d-medium at $T \in \{4, 7\}$, each with seeds 0–1—were selected post hoc from visible stability-frontier transitions, so inference is descriptive of this targeted sample. One shadow uses the first-hop target actor and the other the final-hop target actor. The simulator audit evaluates 12 cloned states at four checkpoints in each of eight runs; inference aggregates checkpoint-level state medians to the training-run level.

For route-native evaluation, ρ_1 and ρ_K are respectively the Polyak first and final target actors with the same TD3 clipped target-smoothing distribution. The common-continuation rerun instead evaluates both shadow critics under ρ_1 on the same sampled states and rollout seeds.

The compute-matched study uses the same eight final cells. Sequential $K = 3$ performs actor updates anchored at μ_0, μ_1 , and μ_2 ; the fixed-reference control anchors both later actors at μ_1 ; the direct-final control applies two optimizer calls to one $2T/3$ objective anchored at μ_1 . Every condition performs three actor optimizer calls per delayed update. The target-lag control instead keeps $K = 1$ and changes only the actor-target Polyak rate from .005 to .005/3, leaving the critic-target rate unchanged.

The frozen-critic small-step audit uses the final TD3+BC $T = 1$ critic for all nine environments and two seeds. At 512 interior dataset actions per run it compares the projected explicit step with a projected fixed-point solution of the implicit action equation for $h \in \{.001, .003, .01, .03, .1\}$. This is an action-level local-consistency test, not a return or coupled-training experiment.

B.3 Estimands and uncertainty

The primary high-budget estimand uses the upper seven values of the ordered fourteen-point grid, $T \geq 4$. Within each task it averages the available seeds and those seven budgets, then weights the nine task means equally. This is an operational stress-region aggregate, not a claim of pointwise monotonicity in T or task. A seed-mean environment–budget cell is called collapsed when its normalized score is below 20. Thresholds 0, 10, 20, 30, and 40 expose sensitivity to that descriptive boundary.

For score contrasts we resample the nine task-level high-budget means with replacement for 100,000 draws using analysis seed 20260830. Raw-run collapse risk is first averaged within each task and then resampled by task using the same draw count and seed. These 95% percentile intervals preserve the complete within-task budget and seed structure but do not remove dependence among the three datasets sharing one dynamics family. With only three families, family analyses are sensitivity checks rather than a three-cluster bootstrap. Cross- K means estimate each randomized procedure’s marginal performance. Identical critic initialization across depth is not required for validity; common random numbers would reduce variance. Paired rescue counts remain descriptive under the realized seed coupling.

C ADDITIONAL RESULTS

Beyond the main-text P4–P3 contrast, P4 minus TD3+BC gains 38.59 points with task-resampling interval [21.39, 54.63] and eight of nine positive task means. At the score-20 threshold, the P4–P3 and L3–L2 raw collapse-risk differences are $-.119$ ($[-.218, -.024]$) and $-.067$ ($[-.131, -.004]$). BAR-Lin is positive in 45/63 continuous cell contrasts (median +1.37); its five rescues and one reversal account for 42.8% of the total gain. Thresholded transitions remain descriptive.

Table 2: Collapse-count sensitivity for $T \geq 4$. Entries are seed-mean cells / raw seeded runs, with denominators 63 / 252 for every method.

Threshold	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
0	1/20	0/14	0/13	0/10	1/17	2/23
10	30/134	22/103	15/81	6/50	24/105	17/85
20	34/142	22/111	16/84	8/54	25/112	21/95
30	36/149	26/115	18/88	11/55	29/120	24/99
40	38/165	30/120	19/96	14/58	32/125	24/105

Table 3: Per-environment high-budget ($T \geq 4$) means.

Environment	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
Hopper-M	11.1	28.7	63.7	84.7	28.0	54.0
Hopper-MR	45.6	92.3	99.6	99.6	93.0	99.0
Hopper-E	3.1	16.8	28.2	46.4	20.8	27.3
HalfCheetah-M	58.8	58.9	58.9	58.2	53.2	55.8
HalfCheetah-MR	51.0	51.8	51.8	52.6	46.3	48.8
HalfCheetah-E	19.9	23.3	30.5	32.8	20.6	27.6
Walker2d-M	9.3	26.0	37.2	59.8	9.0	24.9
Walker2d-MR	27.3	51.2	72.4	76.8	53.4	52.3
Walker2d-E	2.2	21.8	25.1	65.1	21.6	28.2

A restricted four-environment seed-2/3 archive (80 P2/P3 runs) reproduces the reference-critic split: all 45 stable gains are positive and all 35 collapsed gains negative (medians +1.19 and -4.20). It is a sensitivity check of the same implementation, not an independent replication.

Both proximal seed pairs reproduce the strict depth ordering in Table 5. The linearized transition also repeats, with gains of 7.23 ([1.99, 12.47]) on seeds 0–1 and 8.80 ([4.01, 15.21]) on seeds 2–3. These are seed replications of one implementation.

The paired route-only study uses the same eight runs and aggregates four checkpoints per run. For critic route $r \in \{\text{first}, \text{final}\}$ and evaluation continuation ρ , define the checkpoint state-median symmetric relative error

$$E_r^\rho = \text{median}_s \frac{|Q_r(s, a_K) - Q_{\text{MC}}^\rho(s, a_K)|}{|Q_r(s, a_K)| + |Q_{\text{MC}}^\rho(s, a_K)| + 10^{-12}}. \quad (13)$$

The route-native run statistic averages $E_{\text{final}}^{\rho_K} - E_{\text{first}}^{\rho_1}$ over four checkpoints: final-route error is larger in all four-checkpoint aggregates but only five of eight final checkpoints. The run-level mean is .124, median .013, exact two-sided sign-flip $p = .0078$, and run-bootstrap mean 95% interval [.008, .323]. Because this native comparison changes both critic route and continuation, it is not a common-value comparison. We therefore reran both critics under ρ_1 . The common-continuation difference is positive in six of eight runs, with mean .116, median .0022, sign-flip $p = .164$, and bootstrap interval [.0003, .317]. The pre-specified support rule requires a positive mean and two-sided exact sign-flip $p < .05$; the common-continuation result fails the second condition. Neither comparison estimates a return effect because the shadows do not feed back into actor training.

In the frozen-critic audit, 15 of 18 environment–seed runs have an eligible converged, projection-free point. Every eligible run has maximum local relative explicit–implicit discrepancy below 2.7%, with median run maximum .55%. The remaining three have no eligible point. This supports local numerical agreement, not a universal convergence rate.

C.1 Direct target exposure and simulator-reference screening

The original direct audit covers 270 method checkpoints: TD3+BC, P2, and P3 over nine environments, five budgets, and seeds 0–1. A second dump covers all 180 P4 checkpoints over the same environments and budgets with seeds 0–3. P4 comparisons to TD3/P3 use the 90 matched seed-0/1 cells. Table 7 separates critic-target exposure from final-policy reach. Ratios are descriptive sample-anchored squared-displacement ratios, not conditional-mean policy distances.

Table 4: Core matched $K = 2$ diagnostic medians over $T \in \{4, 7, 10, 14, 20\}$. Stable and collapsed groups are defined by external return, while the remaining rows diagnose the learned trajectory.

	Lin st.	Lin col.	Prox st.	Prox col.
No. runs	52	38	59	31
TD p99	270	9.6×10^{23}	160.8	5.0×10^{21}
$\hat{D}_{\text{critic}}$	.112	.538	.064	.446
$\hat{D}_{\text{final}}$	.147	.626	.094	.578
Common- Q gain	+2.11	-1.42	+2.51	-1.06

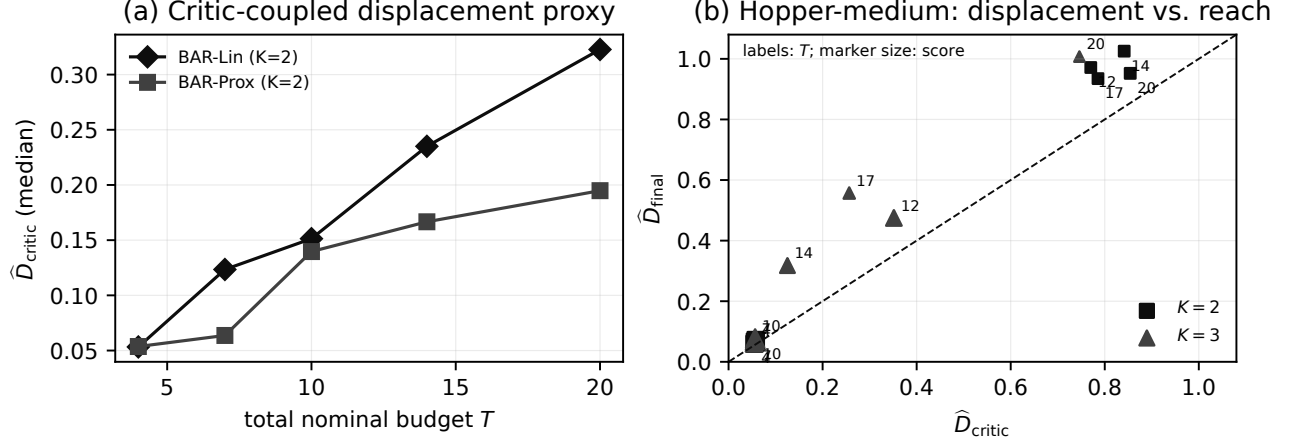


Figure 1: Realized movement. (a) The proximal realization produces a smaller current-state first-hop displacement proxy for $T \geq 7$. (b) On Hopper-medium, $K = 3$ either increases final reach at approximately fixed displacement or contracts both displacements in the observed score-transition region. The displayed current-state quantities exclude Polyak target lag and smoothing noise. Labels indicate T and marker size is proportional to score.

A restricted four-environment seed-2/3 extension reproduces the P3/P2 target exposure direction in 36/40 cells (median ratio .905). It is a post-hoc sensitivity check, not a new full-grid estimand.

The full P4 dump records the target actor and a current-state first-branch proxy, not the final actor. On the 24 overlapping frontier cells with saved final-actor geometry, P4 has lower final displacement in 13/24 comparisons with TD3+BC (median ratio .468) and 5/24 with P3 (1.062). Across all 180 P4 cells, target-displacement medians are .051 for 144 stable runs and .130 for 36 collapsed runs. Thus the 90-cell target contraction does not establish universal final-policy contraction.

For the final-policy proxy, the median P3/TD3 ratio is 1.057, whereas the arithmetic mean difference is $-.104$ with task-seed-cluster interval $[-.194, -.026]$; a few large contractions dominate the mean.

The current-state proxy and deterministic target exposure have Pearson correlation .9999 over all 270 cells; their median ratio is .997 and the largest cellwise relative difference is 5.7%. The corresponding correlation after target smoothing is .9993. The close alignment is an empirical property of these checkpoints, where Polyak lag is small, rather than a mathematical equivalence. Among the 84 Prox-3/Prox-2 runs in which Prox-3 lowers deterministic target exposure, Prox-3 is nevertheless worse by more than one score point in 17. This rejects the stronger interpretation that minimizing exposure alone determines performance.

Across 55 stable and five collapsed checkpoints (720 states), the median checkpoint-level signed critic errors are -41.4 and $+1.70 \times 10^{12}$; 58/60 episode-cluster intervals exclude zero. Restoring episode clocks changes 21 checkpoint medians but preserves all 60 signs (overall median -36.4). The action-ranking subset is too sparse for a population claim: only 69/720 states clear the dead zone for both actions, with 29 sign disagreements.

Table 5: Host-separated full-grid replication. High is the $T \geq 4$ mean; Cell is the number of seed-pair-mean cells below 20 out of 63.

K	High 0–1	High 2–3	Cell 0–1	Cell 2–3
1	25.97	24.79	35	35
2	42.02	40.38	25	26
3	52.78	51.05	17	16
4	63.61	64.33	9	10

Table 6: Targeted final-return controls on the same eight environment–budget–seed cells. The three $K = 3$ rows are compute matched to three actor optimizer calls; the $K = 1$ rows differ only in actor-target Polyak rate.

Condition	Mean	Median	Score < 20
Sequential $K = 3$	71.4	88.9	2/8
Fixed-ref. $K = 3$	70.0	87.9	2/8
Direct $K = 3$	59.4	89.6	3/8
Standard $K = 1$	13.6	0.8	6/8
3× target lag, $K = 1$	12.0	1.5	7/8

Table 7: Matched-cell audit of deterministic next-state Polyak-target exposure and current-state final-policy displacement. The Lower column counts runs in which the left method has the smaller quantity.

Comparison	Quantity	Lower	Ratio
P3 / TD3	Target disp.	88/90	.658
P3 / TD3	Final disp.	36/90	1.057
P3 / P2	Target disp.	84/90	.903
P4 / TD3	Target disp.	89/90	.568
P4 / P3	Target disp.	85/90	.950